Rough Heston with jumps - joint calibration to SPX/VIX level and skew as $T \to 0$

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Abstract

We augment the well known rough Heston model with an additional independent CGMY-type jump process with a Brownian component, and we show that one can simultaneously use the rough Heston parameters to fit the at-the-money VIX level and skew as $T \to 0$, and the CGMY parameters to fit the observed level, at-the-money correction and at-the-money skew of SPX options as $T \to 0$ (using the main Theorem in [FSV21] adapted for our rough Heston V process), and the drift of the V process can be made to be fully consistent with the initial observed variance curve structure. In this sense, the additional CGMY component allows the SPX and VIX smiles to decouple as $T \to 0$ in some sense (which is not possible for a rough model without jumps), and the model can generate power-law skew for both smiles with different effective H-values.

1 Introduction

The Rough Heston stochastic volatility model was introduced in Jaisson&Rosenbaum[JR16], and (using C-tightness arguments from Jacod&Shiryaev[JS13]) they show that the model arises naturally as a weak large-time limit of a high-frequency market microstructure model driven by two nearly unstable Hawkes processes. [ER19] show that the characteristic function of the log stock price for the Rough Heston model admits a quasi-closed form solution via the solution to a non-linear Volterra integral equation (VIE) (see also [ER18]), and the variance curve for the model evolves as $d\xi_t(u) = \kappa(u-t)\sqrt{V_t}dW_t$, where $\kappa(t)$ is the usual fractional kernel $t^{H-\frac{1}{2}}$ for the V process multiplied by a *Mittag-Leffler* function. The instantaneous variance process V for the model is $(H-\varepsilon)$-Hölder continuous like fractional Brownian motion (see e.g. Theorem 3.2 in [JR16]) and the model exhibits power law skew in the small-time limit (see Theorem 3.1 in [FGS21] and Corollary 3.4 in [FSV21]). The theoretical value of the VIX index at time t is $\text{VIX}_t = \sqrt{-\frac{2}{\Delta}\mathbb{E}(\log \frac{S_{t+\Delta}}{S_t}|\mathcal{F}_t)}$, where S_t is the S&P 500 index value at time t, $\Delta = 30$ days and $\mathcal{F}_t$ is the market filtration, so VIX_t is effectively a rolling 30-day Variance swap rate. A VIX option is a European call or put option on VIX_T for some maturity T, and if we replace the spot value S_0 in the Black-Scholes formula with the VIX future price $\mathbb{E}(\text{VIX}_T)$, we can define the implied volatility of a VIX call or put in the usual way by inverting the Black-Scholes formula. VIX options are very liquid in practice (although their bid/offer spreads are still comparatively high), and empirical and model-generated VIX implied volatility smiles have always exhibited a marked positive skew.

In [FGS21b], we consider a generalized Rough Heston model with non-zero initial variance curve $\xi_0(t)$, and we show that (modulo a scaling factor) $(\text{VIX}_T^2 - \text{VIX}_0^2)/T^{\frac{1}{2}-H}$ satisfies the same small-time LDP as the re-scaled log stock price $(X_T - X_0)/T^{\frac{1}{2}-H}$ in the main Theorem 3.3 in [FGS21] if we set $\rho = 1$. We later translate this LDP into VIX call option and implied volatility asymptotics, and we compute a small log-moneyness expansion for the asymptotic VIX smile using expansions previously derived in [FGS21] which yields tractable expressions for the overall level, skew and convexity of the short-end VIX smile.

Unfortunately, since the limiting VIX smile only depends on the factor $\nu/\sqrt{V_0}$ and not on ρ, we cannot simultaneously fit the overall level and skew of observed limiting VIX smile using the standard rough Heston model. To circumvent this issue, we enrich the model in this article with an additional independent CGMY (a.k.a. KoBoL)-type Lévy process L as in [FSV21] with $Y \in (1,2)$ (which implies L has infinite variation), and using a simple modification of the main result in [FSV21] for the Edgeworth regime where log-moneyness scales like $x\sqrt{T}$, we show that the SPX and VIX smiles decouple in some sense as $T \to 0$ if $1 - Y/2 < H$.

More specifically, we can use the rough Heston parameters H, V_0, ν and the initial value VIX_0 of the VIX to fit the overall level and the at-the-money skew of VIX options as $T \to 0$. We can then use the Brownian component σ and the CGMY jump parameters Y, C_+ and C_- to fit the observed at-the-money implied volatility as $T \to 0$, the first order at-the-money correction and the at-the-money skew of SPX options as $T \to 0$. Finally we choose the CGMY parameters G and/or M and $\xi_0(t) = \mathbb{E}(V_t)$ in the definition of the V process to be consistent with the observed variance term structure and the value that we have just calibrated for V_0. The upshot of this is that our CGMY-rough Heston model can generate different power-law skew for the SPX and VIX smiles in the $T \to 0$ limit.

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2 The model

Consider a generalized Rough Heston model for a log stock price process $X_t = \log S_t$:

$$\left\{ \begin{array}{rcl} dX_t &=& -\frac{1}{2}V_t dt + \sqrt{V_t}(\rho dW_t + \bar{\rho} dB_t)\,, \\ V_t &=& \xi_0(t) + c_\alpha \int_0^t (t-s)^{\alpha-1} \nu \sqrt{V_s} dW_s \end{array} \right. \tag{1}$$

for $H \in (0, \frac{1}{2})$, $\alpha = H + \frac{1}{2}$, $c_\alpha = \frac{1}{\Gamma(\alpha)}$ and $\nu > 0$, with some initial variance curve $\xi_0(t)$ with $\xi_0(.)$ continuous, where W, B are two independent Brownian motions, $\bar{\rho} = \sqrt{1-\rho^2}$ with $|\rho| \leq 1$, and we assume $X_0 = 0$ and zero interest rate without loss of generality. It is not known whether we have pathwise uniqueness for (1) even when $\xi_0(t)$ is constant because $\sqrt{v}$ is not Lipschitz at zero (see section 4.2.3 in [JP20] for more on this), but we do have weak uniqueness (see Theorem 3.4 in [ALP19]) and uniqueness in law for V on $C([0,T])$, since we can explicitly compute an exponential-affine formula for the Fourier transform of V on pathspace in terms of a Volterra integral equation with a unique solution, see Theorem 7.1 in [ALP19] (see also Theorem 6.1 there); the fixed-t version of this formula is used in Appendix B.

2.1 Adding jumps - decoupling the SPX and VIX short-maturity skews

We now augment our rough Heston model to include jumps of infinite variation. Specifically (following [FSV21]), we now assume that the log stock price process X evolves as

$$dX_t \;=\; -\frac{1}{2}V_t dt + \sqrt{V_t}(\rho dW_t + \bar{\rho} dB_t) - dL_t \tag{2}$$

where L is a Lévy process independent of W and B with $\mathbb{E}(e^{-uL_t}) = e^{tV(u)}$ for u in some interval (u_-, u_+) which includes 0 and 1 with $\mathbb{E}(e^{-L_t}) = 1$ (i.e. $V(1) = 0$) which ensures that e^{-L} is a martingale with respect to its own filtration, and we assume that $X_0 = 0$ and V is not equal to the zero function W.L.O.G.

Then

$$-\frac{2}{\Delta}\mathbb{E}(X_{T+\Delta} - X_T|\mathcal{F}_T) \;=\; \frac{1}{\Delta}\mathbb{E}(\int_T^{T+\Delta} V_u du|\mathcal{F}_T) - \frac{2}{\Delta}\mathbb{E}(-(L_{T+\Delta} - L_T)|\mathcal{F}_T) \;=\; \frac{1}{\Delta}\int_T^{T+\Delta} \xi_T(u) du + a_1$$

where $\xi_t(u) := \mathbb{E}(V_u|\mathcal{F}_t)$, and

$$a_1 \;:=\; -2V'(0) \;=\; -\frac{2}{\Delta}\mathbb{E}(-(L_{T+\Delta} - L_T)|\mathcal{F}_T) \;\geq\; -\frac{2}{\Delta}\log \mathbb{E}(e^{-(L_{T+\Delta} - L_T)}|\mathcal{F}_T) \;=\; -\frac{2}{\Delta}\log 1 \;=\; 0 \tag{3}$$

and we have used Jensen and the martingale condition to obtain the inequality here. Moreover, since $V(0) = V(1) = 0$, $V'(0) < 0$ (since we are assuming that V is not identically zero), so we see that a_1 is strictly positive.

Then we see that $\mathrm{VIX}_T^2 = \frac{1}{\Delta}\int_T^{T+\Delta} \xi_T(u) du + a_1$, so the effect of the jumps is to increase the value of VIX_T^2 by a *shift factor*[1] equal to a_1. Thus $(\mathrm{VIX}_T^2 - a_1 - \frac{1}{\Delta}\int_0^\Delta \xi_0(u) du)/T^{\frac{1}{2}-H} = (\mathrm{VIX}_T^2 - \mathrm{VIX}_0^2)/T^{\frac{1}{2}-H}$ now satisfies the LDP as $T \to 0$ with speed T^{-2H} and rate function $J(x)$ as before, and we have the following:

Corollary 2.1

$$\begin{array}{rcl} \lim\limits_{T\to 0} T^{2H} \log \mathbb{E}((\mathrm{VIX}_T - (a_1 + \frac{1}{\Delta}\int_0^\Delta \xi_0(u) du)^{\frac{1}{2}}\, e^{x T^{\frac{1}{2}-H}})_+) &=& \lim\limits_{T\to 0} T^{2H} \log \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\, e^{x T^{\frac{1}{2}-H}})_+) \\ &=& -J(2(a_1 + \frac{1}{\Delta}\int_0^\Delta \xi_0(u) du)x) \\ &=& -J(2\mathrm{VIX}_0^2 x) \end{array}$$

for $x > 0$, where $J(.)$ is the same as in the main Theorem 2.1 in [FGS21b].

Proof. See Appendix A. ■

Corollary 2.2

$$\hat{\sigma}_{\mathrm{VIX}}(x) \;:=\; \lim_{T\to 0} \hat{\sigma}_{\mathrm{VIX}}(\mathrm{VIX}_0 e^{x T^{\frac{1}{2}-H}}, T) \;=\; \frac{|x|}{\sqrt{2J(2\mathrm{VIX}_0^2 x)}}\,. \tag{4}$$

[1] Note this is conceptually similar to a simple displaced-diffusion model $dS_t = (\beta S_t + 1 - \beta)dW_t$ for S with $\beta \in (0,1)$ where S can go negative, but here we do not have this negativity problem since $a_1 \geq 0$. This also means that a VIX put option with $K \leq \sqrt{a_1}$ is worthless.

Expanding the final expression in Corollary 2.1, we find that

$$\begin{aligned}\hat{\sigma}_{\text{VIX}}(x) &= \frac{\nu\sqrt{V_0}}{\frac{1}{\Delta}\int_0^\Delta \xi_0(u)du + a_1}\frac{\Delta^{\alpha-1}}{2\alpha\Gamma(\alpha)} + \frac{\nu}{2\sqrt{V_0}\Gamma(2+\alpha)}x + O(x^2) \\ &= \frac{\nu\sqrt{V_0}}{\text{VIX}_0^2}\frac{\Delta^{\alpha-1}}{2\alpha\Gamma(\alpha)} + \frac{\nu}{2\sqrt{V_0}\Gamma(2+\alpha)}x + O(x^2)\,. \end{aligned} \tag{5}$$

Remark 2.1 Since $a_1 \geq 0$, we see that if the leading order term is fixed, adding jumps can only increase the skew term relative to the leading order term.

2.2 CGMY jumps and asymptotics for the SPX skew as $T \to 0$ in the Edgeworth regime

From here on we consider the special case when L is a generalized tempered stable (i.e. CGMY-type) process with $L_0 = 0$ and Lévy triple $(b, \sigma^2, \nu(x) = \frac{C_+e^{-Mx}}{x^{1+Y}}1_{x>0} + \frac{C_-e^{-G|x|}}{|x|^{1+Y}}1_{x<0})$ with $Y \in (1,2)$, which implies that

$$\phi_t^L(u) := \mathbb{E}(e^{-iu(L_t-L_0)}) = e^{t\psi(u)} = e^{iubt - \frac{1}{2}\sigma^2u^2t + C_+\Gamma(-Y)t[(M+iu)^Y - M^Y] + C_-\Gamma(-Y)t[(G-iu)^Y - G^Y]} \tag{6}$$

and $\mathbb{E}(e^{-pL_t}) = e^{V(p)t}$ for $p \in (-M, G)$, where $V(p) = \psi(-ip)$. Note that b is fixed by the martingale condition $V(1) = 0$.

Remark 2.2 Corollary 2.10 and Remark 2.7 in [FSV21] describe a method for simulating L_t by changing to a measure under which L is a pure α-stable process, which allows for long-time stepping using the classical Chambers,Mallows&Stuck[CMS76] formula for sampling stable random variables.

We can replace the Y process in the main Theorem 2.1 in [FSV21] with our rough Heston V process (since the only special feature of the Y process required in [FSV21] is bivariate weak convergence of $(\frac{1}{\sqrt{t}}\hat{X}_t, (Y_t - \mathbb{E}(Y_t))/t^H)$ as $t \to 0$ to a joint Gaussian under Eq 17 in [FSV21], which is exact there even for $t > 0$ (but this is not required) but is also established in Appendix B of this article for $(\frac{1}{\sqrt{t}}\hat{X}_t, (V_t - \mathbb{E}(V_t))/t^H)$), and as in section 2.1 in [FSV21] we can extend from the case of bounded volatility to the true driftless rough Heston model (see Appendix C for details) and our S process is a martingale and the history term ζ in our context here (see just below Eq 5 in [FSV21]) is given by $\zeta(t) = \mathbb{E}(V_t) - V_0 = \xi_0(t) - V_0$ in our setup here, and the contribution from this term is contained in the error term in (7) if $\xi_0(t) - V_0 = O(t^H)$ as $t \to 0$. Thus Theorem 2.1 (and Lemma 2.7) in [FSV21] still apply for the model in (2), which we now recall:

Proposition 2.3 *Let $\hat{\sigma}(K,T)$ denote the implied volatility of a European call option with strike K and maturity T at time zero. For the model in (2), if $H \in (1 - \frac{1}{2}Y, \frac{1}{2})$, $G > 1$ and $M > 1$, we have the following asymptotic behaviour for implied volatility at time zero in the Edgeworth regime:*

$$\hat{\sigma}(e^{z\sqrt{T}}, T) = \sqrt{V_0 + \sigma^2} + \frac{1}{\phi(\frac{z}{\sqrt{V_0+\sigma^2}})}(C_+A_+(z) + C_-A_-(z))T^{1-\frac{1}{2}Y} + o(T^{1-\frac{1}{2}Y}) \tag{7}$$

for $z \in \mathbb{R}$ as $T \to 0$ and

$$A_\pm(z) = \frac{1}{\pi}\int_0^\infty \text{Re}[e^{-iuz}e^{-\frac{1}{2}(V_0+\sigma^2)u^2}\Gamma(-Y)(\pm iu)^{Y-2}]du$$

where ϕ is the standard Normal density.

Remark 2.3 $\frac{1}{\phi(\frac{z}{\sqrt{V_0+\sigma^2}})}(C_+A_+(z) + C_-A_-(z))$ is non-linear in z and does not vanish at $z = 0$ in general (see e.g. Figure 6 in [FSV21] for a plot), and C_+ and M relate to negative jumps for X and C_- and G relate to positive jumps, since we are subtracting L in (2). Note that the log stock price X_t does not satisfy a small-time LDP as $t \to 0$ (even if re-scaled).

Remark 2.4 Note that a_1 depends on M and G as well, which do not show up in the leading order Edgeworth asymptotics in (7).

Hence the addition of the CGMY-type jump process L allows the asymptotic SPX and VIX short time smiles to have power-law skews with different H-values since the effective H as $T \to 0$ for the former is now $1 - Y/2$ (due to jumps alone, see (7)) and the latter is $H = \alpha - \frac{1}{2}$ (due to V alone), so (aside from (7) depending on V_0), we have effectively de-coupled the short-time behaviour of the SPX and VIX skews.

2.3 Exact calibration to the at-the-money skew for SPX and VIX as $T \to 0$

We let $\hat{\sigma}^{\rm mkt}_{\rm VIX}(K,T)$ denote the *market* implied volatility of a VIX call option with strike K and maturity T. If there exists an $\alpha = H + \frac{1}{2}$ such that $\lim_{T\to 0} \hat{\sigma}^{\rm mkt}_{\rm VIX}({\rm VIX}_0 e^{xT^{\frac{1}{2}-H}}, T)$ has a non-trivial limit which is differentiable in x, then we can choose V_0 and ν so that the market observed behaviour of VIX options as $T \to 0$ is consistent with (5). We can then choose parameters so that (7) is consistent with the observed behaviour of SPX options in the market. Specifically choose σ to fit the observed behaviour of the limiting at-the-money implied vol $\lim_{T\to 0} \hat{\sigma}^{\rm mkt}(1,T)$, and then solve for C_+, C_- and Y to match the observed behaviour of the at-the-money correction for European options: $\lim_{T\to 0} \frac{\hat{\sigma}^{\rm mkt}(1,T) - \hat{\sigma}^{\rm mkt}(1,0)}{T^{1-Y/2}}$ and the skew term: $\lim_{z\to 0} \lim_{T\to 0} \frac{\hat{\sigma}^{\rm mkt}(e^{z\sqrt{T}},T) - \hat{\sigma}^{\rm mkt}(1,T)}{zT^{1-Y/2}}$, assuming we can find a $Y \in (1,2)$ such that both these limits are finite (and at least one is non-zero). This leads to the following two equations for $C_\pm$:

$$\sqrt{2\pi}(C_+ A_+(0) + C_- A_-(0)) = \lim_{T\to 0} \frac{\hat{\sigma}^{\rm mkt}(1,T) - \hat{\sigma}^{\rm mkt}(1,0)}{T^{1-Y/2}} \tag{8}$$

$$\sqrt{2\pi}(C_+ A'_+(0) + C_- A'_-(0)) = \lim_{z\to 0} \lim_{T\to 0} \frac{\hat{\sigma}^{\rm mkt}(e^{z\sqrt{T}},T) - \hat{\sigma}^{\rm mkt}(1,T)}{zT^{1-Y/2}} \tag{9}$$

obtained by computing $\frac{1}{\phi(\frac{z}{\sqrt{V_0+\sigma^2}})}(C_+ A_+(z) + C_- A_-(z))$ and its derivative at $z = 0$. The last two equations are linear in C_+ and C_- and hence can be solved explicitly if the Wronskian of $A_+(z)$ and $A_-(z)$ is non-zero at $z = 0$, and we can verify that this is indeed the case since

$$\begin{aligned} A_\pm(z) &= \frac{1}{\pi}\int_0^\infty {\rm Re}[e^{-iuz} e^{-\frac{1}{2}(V_0+\sigma^2)u^2} \Gamma(-Y)(\pm iu)^{Y-2}]du \\ &= \frac{1}{\pi}\int_0^\infty {\rm Re}[e^{-iuz} e^{-\frac{1}{2}(V_0+\sigma^2)u^2} \Gamma(-Y) e^{\pm\frac{1}{2}i\pi(Y-2)} u^{Y-2}]du \\ &= \frac{1}{\pi}\int_0^\infty \cos(-uz \pm \frac{1}{2}\pi(Y-2)) e^{-\frac{1}{2}(V_0+\sigma^2)u^2} \Gamma(-Y) u^{Y-2} du \end{aligned}$$

which implies that

$$\begin{aligned} A'_\pm(z) &= \frac{1}{\pi}\int_0^\infty {\rm Re}[u\sin(-uz \pm \frac{1}{2}\pi(Y-2)) e^{-\frac{1}{2}(V_0+\sigma^2)u^2} \Gamma(-Y) u^{Y-2}]du \\ A_\pm(0) &= \frac{1}{\pi}\int_0^\infty {\rm Re}[\cos(\frac{1}{2}\pi(Y-2)) e^{-\frac{1}{2}(V_0+\sigma^2)u^2} \Gamma(-Y) u^{Y-2}]du \\ A'_\pm(0) &= \pm\frac{1}{\pi}\int_0^\infty {\rm Re}[u\sin(\frac{1}{2}\pi(Y-2)) e^{-\frac{1}{2}(V_0+\sigma^2)u^2} \Gamma(-Y) u^{Y-2}]du \end{aligned}$$

i.e. $A_+(0) = A_-(0)$ and $A'_+(0) = -A'_-(0)$.

We also have to make the model consistent with the observed variance term structure (i.e. prices of variance swaps at all maturities implied by European option prices) up to some maturity $T^* \in (0, \Delta]$, for which

$$\begin{aligned} -2\mathbb{E}(\log\frac{S_t}{S_0}) &= \int_0^t \xi_0(u)du + a_1 t \\ \Rightarrow \quad -\frac{\partial}{\partial t} 2\mathbb{E}(\log\frac{S_t}{S_0}) &= \xi_0(t) + a_1 \\ \Rightarrow \quad -\frac{\partial}{\partial t} 2\mathbb{E}(\log\frac{S_t}{S_0})_{t=0} &= \xi_0(0) + a_1 = V_0 + a_1 . \end{aligned} \tag{10}$$

a_1 depends on G and M, which must be chosen so the calibrated value $\xi_0(0) = V_0$ here matches our calibrated value of V_0 from above; then (10) uniquely determines $\xi_0(t)$ for $t > 0$ as well (and this includes being consistent with the market value of ${\rm VIX}_0$ which corresponds to $t = \Delta$), which we have implicitly used above for the VIX smile calibration.

We now discuss the range of admissible values for a_1. Imposing the martingale condition $V(1) = 0$ we find that

$$\begin{aligned} a_1 = a_1(G,M) &= \sigma^2 - 2C_+(M^Y - (1+M)^Y + M^{Y-1}Y)\Gamma(-Y) \\ &+ 2C_-((G-1)^Y - G^Y + G^{Y-1}Y)\Gamma(-Y) . \end{aligned}$$

Then $a_1(1,1) > 0$ and $a_1(G,M) \to \sigma^2$ as $M, G \to \infty$, and we have the following monotonicity result:

Lemma 2.4 *$a_1(G,M)$ is continuous and strictly decreasing in M and G.*

Proof.

$$\frac{\partial a_1}{\partial M} = -2C_+(YM^{Y-1} - Y(1+M)^{Y-1} + Y(Y-1)M^{Y-2})\Gamma(-Y) \tag{11}$$

Now consider the function $g_M(K) = (K+M)^{Y-1}$. This function is concave so

$$(K+M)^{Y-1} \leq M^{Y-1} + (Y-1)M^{Y-2}K$$

Setting $K = 1$ we see that

$$(1+M)^{Y-1} \leq M^{Y-1} + (Y-1)M^{Y-2}$$

and the result follows by comparing this to (11). We use a similar argument for the term involving G. ■

Since Proposition 2.3 requires that $G, M > 1$, we see that there is an interval (σ^2, a_1^*) of admissible a_1-values, where $a_1^* = a_1(1,1) = \sigma^2 + 2(C_+(2^Y - Y - 1) + C_-(Y-1))\Gamma(-Y)$ for Y in the admissible range $Y \in (1,2)$. If the calibrated V_0-value is such that $-\frac{\partial}{\partial t}2\mathbb{E}(\log \frac{S_t}{S_0})_{t=0} - V_0 \notin (\sigma^2, a_1^*)$ (see (10)) then it means the model is mis-specified, and this ultimately puts an upper and lower bound on the admissible ratio of the first to the zeroth order term in (5).

Of course in practice we cannot perfectly extract all the limiting quantities on the right hand side of (8) and (9) from the market since only a finite number of options are traded, and there are real-life issues with noisy data, bid-offer spreads, day count/weekend conventions etc., but one can approximate the quantities on the right with finite differences to incorporate information from options with small but non-zero maturity. These results can also be used to make smart initial guesses for conventional and deep learning-based calibration schemes.

J.Guyon has remarked[2] that fitting large SPX short-end skew and (comparatively) small VIX short-end implied volatility is the main challenge in joint fitting of SPX and VIX smiles, so this gives a concrete result in this direction.

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A Asymptotics for VIX call options

From Jensen's inequality, we know that for any $q \geq 1$ we have

$$(\mathrm{VIX}_T^2)^q \;=\; (a_1 + \frac{1}{\Delta}\int_T^{T+\Delta} \xi_T(u)du)^q \;=\; (\frac{1}{\Delta}\int_T^{T+\Delta}(a_1+\xi_T(u))du)^q \;\leq\; \frac{1}{\Delta}\int_T^{T+\Delta}(a_1+\xi_T(u))^q du$$

and hence

$$\begin{aligned}\mathbb{E}(\mathrm{VIX}_T^{2q}) &\leq \frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}((a_1+\xi_T(u))^q)du \;=\; \frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}(\,\mathbb{E}(a_1+V_u|\mathcal{F}_T)^q)du \\ &\leq \frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}((a_1+V_u)^q)du \qquad \text{(A-1)}\end{aligned}$$

which will be needed further down.

- Lower bound. We first note that for x fixed and any $\delta \in (0,|x|)$, $e^{xT^{\frac{1}{2}-H}} \leq 1+(x+\delta)T^{\frac{1}{2}-H}$ for T sufficiently small. Recall that $\mathrm{VIX}_0^2 = a_1 + \frac{1}{\Delta}\int_0^\Delta \xi_0(u)du$ and we set $k_{x,\delta} := \mathrm{VIX}_0(x+\delta)$. Thus for $T = T(\delta)$ sufficiently small

$$\begin{aligned}\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0 e^{xT^{\frac{1}{2}-H}})_+) &\geq \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0(1+(x+\delta)T^{\frac{1}{2}-H})_+) \\ &= T^{\frac{1}{2}-H}\mathbb{E}((\frac{\mathrm{VIX}_T-\mathrm{VIX}_0}{T^{\frac{1}{2}-H}} - k_{x,\delta})_+) \\ &\geq \delta T^{\frac{1}{2}-H}\mathbb{E}(1_{\frac{\mathrm{VIX}_T-\mathrm{VIX}_0}{T^{\frac{1}{2}-H}} > k_{x,\delta}+\delta}) \\ &= \delta T^{\frac{1}{2}-H}\mathbb{P}(\mathrm{VIX}_T > \mathrm{VIX}_0 + T^{\frac{1}{2}-H}(k_{x,\delta}+\delta)) \\ &= \delta T^{\frac{1}{2}-H}\,\mathbb{P}(\mathrm{VIX}_T^2 > \mathrm{VIX}_0^2 + 2\mathrm{VIX}_0\,(k_{x,\delta}+\delta)T^{\frac{1}{2}-H} + (k_{x,\delta}+\delta)^2 T^{1-2H})\,.\end{aligned}$$

  But for $T = T(\delta)$ sufficiently small, the right hand side here is greater than or equal to

$$\begin{aligned}&\delta T^{\frac{1}{2}-H}\,\mathbb{P}(\mathrm{VIX}_T^2 - \mathrm{VIX}_0^2 > 2\mathrm{VIX}_0\,(k_{x,\delta}+2\delta)T^{\frac{1}{2}-H}) \\ =\;& \delta T^{\frac{1}{2}-H}\,\mathbb{P}(\frac{1}{\Delta}\int_T^{T+\Delta}\xi_T(u)du - \frac{1}{\Delta}\int_0^\Delta \xi_0(u)du > 2\mathrm{VIX}_0\,(k_{x,\delta}+2\delta)T^{\frac{1}{2}-H})\,.\end{aligned}$$

  Then using the LDP and the continuity of J we see that

$$\liminf_{T\to 0} T^{2H}\log\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\,e^{xT^{\frac{1}{2}-H}})_+) \;\geq\; -J(2\mathrm{VIX}_0(k_{x,\delta}+2\delta)) \;=\; -J(2\mathrm{VIX}_0^2 x + 2\delta\mathrm{VIX}_0^2 + 4\delta\mathrm{VIX}_0)\,.$$

  We then let $\delta \to 0$ and again use the continuity of the rate function $J(x)$ to obtain the required lower bound.

- Upper bound. From Hölder's inequality, we note that for $q > 1$

$$\begin{aligned}\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\,e^{xT^{\frac{1}{2}-H}})_+) &\leq \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H}))_+) \\ &= \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H}))_+ 1_{\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H})}) \\ &\leq \mathbb{E}[(\mathrm{VIX}_T - \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H}))_+^q]^{\frac{1}{q}}\,\mathbb{E}(1_{\mathrm{VIX}_T \geq \mathrm{VIX}_0(1+xT^{\frac{1}{2}-H})})^{1-\frac{1}{q}}\,.\end{aligned}$$

  Thus

$$\begin{aligned}&T^{2H}\log\mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))_+) \\ \leq\;& \frac{T^{2H}}{q}\log\mathbb{E}[(\mathrm{VIX}_T - \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H}))_+^q] \;+\; T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H})) \\ \leq\;& \frac{T^{2H}}{q}\log\mathbb{E}(\mathrm{VIX}_T^q) \;+\; T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H})) \\ \leq\;& \frac{T^{2H}}{q}\log(\mathbb{E}(\mathrm{VIX}_T^{2q})^{\frac{1}{2}}) \;+\; T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T \geq \mathrm{VIX}_0\,(1+xT^{\frac{1}{2}-H})) \\ \leq\;& \frac{T^{2H}}{q}\frac{1}{2}\log(\frac{1}{\Delta}\int_T^{T+\Delta}\mathbb{E}((a_1+V_u)^q)du) \;+\; T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T^2 \geq \mathrm{VIX}_0^2(1+xT^{\frac{1}{2}-H})^2) \\ &\text{(by (A-1))} \\ \leq\;& \frac{T^{2H}}{q}\frac{1}{2}\log(\frac{1}{\Delta}\int_T^{T+\Delta}(a_1+\mathbb{E}(V_u^q)^{\frac{1}{q}})^q du) \;+\; T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T^2 \geq \mathrm{VIX}_0^2(1+2xT^{\frac{1}{2}-H})) \\ &\text{(using Minkowski applied to } \mathbb{E}((a_1+V_u)^q)) \\ \leq\;& \frac{T^{2H}}{q}\frac{1}{2}\log(a_1 + c_{q,T}^{\frac{1}{q}})^q \;+\; T^{2H}(1-\frac{1}{q})\log\mathbb{P}(\mathrm{VIX}_T^2 \geq \mathrm{VIX}_0^2(1+2xT^{\frac{1}{2}-H}))\end{aligned}$$

for some finite constant $c_{q,T}$ depending on q and T, where we have used Lemma C.1 in [FGS21] in the final line. Letting $T \to 0$ in the final line and using the LDP and the continuity of J, and then letting $q \to \infty$, we see that

$$\limsup_{T\to 0} T^{2H} \log \mathbb{E}((\mathrm{VIX}_T - \mathrm{VIX}_0\,(1 + xT^{\frac{1}{2}-H}))_+) \quad \leq \quad -J(2\mathrm{VIX}_0^2 x)\,.$$

B Bivariate weak convergence

Proposition B.1 *Let $\hat{X}_t := \sqrt{V_0}(\rho W_t + \bar{\rho} B_t)$. Then $(X_t/\sqrt{t}, (V_t - \xi_0(t))/t^H)$ and $(\hat{X}_t/\sqrt{t}, (V_t - \xi_0(t))/t^H)$ both tend weakly to the same bivariate Gaussian as $t \to 0$.*

Proof.

$$\mathbb{E}(e^{pX_t + u(V_t - \xi_0(t))}) \quad = \quad e^{\int_0^t \xi_0(t-s)(\frac{1}{2}p^2 - \frac{1}{2}p + p\rho\nu\psi(p,u,s) + \frac{1}{2}\nu^2\psi(p,u,s)^2)ds}$$

for $t \in [0, T^*_\psi(p,u))$, where $\psi(p,u,t)$ satisfies

$$\psi(p,u,t) \quad = \quad uc_\alpha t^{\alpha-1} + \int_0^t c_\alpha (t-s)^{\alpha-1}(\frac{1}{2}p^2 - \frac{1}{2}p + p\rho\nu\psi(p,u,s) + \frac{1}{2}\nu^2\psi(p,u,s)^2)ds$$

and $T^*_\psi(p,u) > 0$ is the explosion time for ψ. Then

$$\mathbb{E}(e^{\frac{p}{\sqrt{\varepsilon}}X_{\varepsilon t} + \frac{u}{\varepsilon^H}(V_{\varepsilon t} - \xi_0(\varepsilon t))}) \quad = \quad e^{\int_0^{\varepsilon t} \xi_0(\varepsilon t - s)(\frac{1}{2}\frac{p^2}{\varepsilon} - \frac{1}{2}\frac{p}{\sqrt{\varepsilon}} + \frac{p}{\sqrt{\varepsilon}}\rho\nu\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, s) + \frac{1}{2}\nu^2\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, s)^2)ds}$$

and

$$\begin{aligned}
\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, \varepsilon t) &= \frac{u}{\varepsilon^H} c_\alpha t^{\alpha-1}\varepsilon^{\alpha-1} + \int_0^{\varepsilon t} c_\alpha(\varepsilon t - s)^{\alpha-1}(\frac{1}{2}\frac{p^2}{\varepsilon} - \frac{1}{2}\frac{p}{\sqrt{\varepsilon}} + \frac{p}{\sqrt{\varepsilon}}\rho\nu\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, s) + \frac{1}{2}\nu^2\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, s)^2)ds \\
&= \frac{u}{\varepsilon^H} c_\alpha t^{\alpha-1}\varepsilon^{\alpha-1} + \varepsilon\int_0^t c_\alpha(\varepsilon t - \varepsilon s)^{\alpha-1}(\frac{1}{2}\frac{p^2}{\varepsilon} - \frac{1}{2}\frac{p}{\sqrt{\varepsilon}} + \frac{p}{\sqrt{\varepsilon}}\rho\nu\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, \varepsilon s) + \frac{1}{2}\nu^2\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, \varepsilon s)^2)ds\,.
\end{aligned}$$

for $t \in [0, \frac{1}{\varepsilon}T^*_\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}))$. Then $\psi^\varepsilon(p,u,t) = \sqrt{\varepsilon}\,\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, \varepsilon t)$ satisfies

$$\psi^\varepsilon(p,u,t) \quad = \quad uc_\alpha t^{\alpha-1} + \int_0^t c_\alpha(t-s)^{\alpha-1}(\frac{1}{2}\varepsilon^H p^2 - \frac{1}{2}p\varepsilon^{\frac{1}{2}+H} + p\varepsilon^H\rho\nu\psi^\varepsilon(p,u,s) + \frac{1}{2}\nu^2\varepsilon^H\psi^\varepsilon(p,u,s)^2)ds$$

for $t \in [0, \frac{1}{\varepsilon}T^*_\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}))$, and

$$\begin{aligned}
\mathbb{E}(e^{\frac{p}{\sqrt{\varepsilon}}X_{\varepsilon t} + \frac{u}{\varepsilon^H}(V_{\varepsilon t} - \xi_0(\varepsilon t))}) &= e^{\int_0^{\varepsilon t} \xi_0(\varepsilon t - s)(\frac{1}{2}\frac{p^2}{\varepsilon} - \frac{1}{2}\frac{p}{\sqrt{\varepsilon}} + \frac{p}{\sqrt{\varepsilon}}\rho\nu\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, s) + \frac{1}{2}\nu^2\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, s)^2)ds} \\
&= e^{\varepsilon\int_0^t \xi_0(\varepsilon t - \varepsilon s)(\frac{1}{2}\frac{p^2}{\varepsilon} - \frac{1}{2}\frac{p}{\sqrt{\varepsilon}} + \frac{p}{\sqrt{\varepsilon}}\rho\nu\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, \varepsilon s) + \frac{1}{2}\nu^2\psi(\frac{p}{\sqrt{\varepsilon}}, \frac{u}{\varepsilon^H}, \varepsilon s)^2)ds} \\
&= e^{\int_0^t \xi_0(\varepsilon t - \varepsilon s)(\frac{1}{2}p^2 - \frac{1}{2}p\sqrt{\varepsilon} + p\rho\nu\psi^\varepsilon(p,u,s) + \frac{1}{2}\nu^2\psi^\varepsilon(p,u,s)^2)ds}\,.
\end{aligned}$$

From Theorem 13.1.1 i) in [GLS90] (see the paragraph above Eq 12 in [FGS21] for details on the use of part i) of Theorem 13.1.1 in [GLS90]), we know that $\psi^\varepsilon(p,u,t)$ tends uniformly to $uc_\alpha t^{\alpha-1}$ as $\varepsilon \to 0$ on any compact interval, and since $\xi_0(.)$ is continuous (and hence bounded on any compact interval) we see that

$$\mathbb{E}(e^{\frac{p}{\sqrt{\varepsilon}}X_{\varepsilon t} + \frac{u}{\varepsilon^H}(V_{\varepsilon t} - \xi_0(\varepsilon t))}) \quad \to \quad e^{V_0(\frac{1}{2}p^2 t + \rho\nu p u\frac{t^\alpha}{\Gamma(\alpha+1)} + \frac{1}{2}\nu^2u^2\frac{t^{2\alpha-1}}{\Gamma(\alpha)^2(2\alpha-1)})}$$

by the bounded convergence theorem. Finally, since Theorem 13.1.1 in [GLS90] is multi-dimensional, we can apply it to $(\mathrm{Re}(\psi), \mathrm{Im}(\psi))$ with p replaced by ik with $k \in \mathbb{R}$ as we do in section 5 in [FGS21]. The result then follows from Lévy's convergence theorem.

Similarly, by Brownian scaling, define $\widetilde{W}^\varepsilon_r := \varepsilon^{-\frac{1}{2}}W_{\varepsilon r}$, $\widetilde{B}^\varepsilon_r := \varepsilon^{-\frac{1}{2}}B_{\varepsilon r}$. Then

$$\begin{aligned}
\frac{\hat{X}_{\varepsilon t}}{\sqrt{\varepsilon}} &= \sqrt{V_0}(\rho\widetilde{W}^\varepsilon_t + \bar{\rho}\widetilde{B}^\varepsilon_t), \\
\frac{V_{\varepsilon t} - \xi_0(\varepsilon t)}{\varepsilon^H} &= \nu c_\alpha\int_0^t (t-r)^{\alpha-1}\sqrt{V_{\varepsilon r}}d\widetilde{W}^\varepsilon_r.
\end{aligned}$$

Since $V_{\varepsilon r} \to V_0$ in probability uniformly for $0 \leq r \leq t$, and $(t-r)^{\alpha-1}$ is square-integrable on this interval, we obtain

$$(\frac{\hat{X}_{\varepsilon t}}{\sqrt{\varepsilon}}, \frac{V_{\varepsilon t} - \xi_0(\varepsilon t)}{\varepsilon^H}) \quad \stackrel{w}{\to} \quad (\sqrt{V_0}(\rho W_t + \bar{\rho}B_t),\, \nu\sqrt{V_0}c_\alpha\int_0^t (t-r)^{\alpha-1}dW_r).$$

The limiting vector is centred Gaussian with moment generating function

$$\exp\{V_0(\frac{1}{2}p^2 t + p\rho\nu u\frac{t^\alpha}{\Gamma(\alpha+1)} + \frac{1}{2}\nu^2u^2\frac{t^{2\alpha-1}}{\Gamma(\alpha)^2(2\alpha-1)})\}.$$

■

C Extending to unbounded volatility

This appendix is a minor variation of Section 2.1 in [FSV21] adapted for our purposes here. We first let $\underline{V}_t := \min_{0\le s\le t} V_s$ and $\bar{V}_t := \max_{0\le s\le t} V_s$, and we consider the stochastic Volterra system

$$\hat{V}_t \;=\; \xi_0(t) \,+\, c_\alpha \nu \int_0^t (t-s)^{H-\frac{1}{2}} \sigma(\hat{V}_s) dW_s$$

where $\sigma \in C_b^2$ as in [FSV21] and bounded away from zero with $\sigma(y) = \sqrt{y}$ for $y \in [a,b]$ for some $a \in (0, V_0)$ and $b > V_0$, so in particular σ is bounded and Lipschitz. Then $V_t = \hat{V}_t$ for $t \in [0, \tau_a \wedge \tau_b)$ where $\tau_a = \inf\{t : V_t = a\}$ (and similarly for τ_b) for any $a \in (0, V_0)$ since we have pathwise uniqueness for $\hat{V}$ (see section 4.2.3 in [JP20] and discussion in section 2 in [FGS21]) and from Proposition 4.3 in [JP20] (or Theorem 3.11 in [Zha08]), we know that $\hat{V}_{t(.)}$ satisfies the large deviation principle on $C_{V_0}[0,1]$ as $t \to 0$ with speed $1/t^{2H}$ and rate function

$$I(\phi) \;=\; \frac{1}{2} \int_0^1 \Big(\frac{D^\alpha(\phi(.) - \phi(0))(t)}{\nu\, \sigma(\phi(t))}\Big)^2 dt$$

if $\phi \in I_{V_0}^{H+\frac{1}{2}}(L^1)$, and $I(\phi) = \infty$ otherwise. Now let $A := \{\underline{V}_t > a\} \cap \{\bar{V}_t < b\}$ and set $\hat{S}_t = e^{\hat{X}_t}$, where

$$d\hat{X}_t \;=\; -\frac{1}{2}\sigma(\hat{V}_t)^2 dt \,+\, \sigma(\hat{V}_t)(\rho dW_t + \bar{\rho} dB_t) \,-\, dL_t$$

and $\hat{X}_0 = 0$ i.e. the same stock price process (defined on the same probability space) but with V_t replaced with $\hat{V}_t$. Then for all $\varepsilon > 0$ we have

$$\begin{aligned}
\mathbb{E}((e^{z\sqrt{t}} - S_t)^+) \;&=\; \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+ 1_A) \,+\, \mathbb{E}((e^{z\sqrt{t}} - S_t)^+ 1_{A^c}) \\
&\le\; \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+) \,+\, e^{z\sqrt{t}} \mathbb{P}(A^c) \\
&\le\; \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+) \,+\, e^{z\sqrt{t}} e^{-\frac{1}{t^{2H}}(-\varepsilon + J(a,b))}
\end{aligned}$$

for t sufficiently small, where $J(a,b) = \inf_{\phi \in C_{V_0}([0,1]) : \underline{\phi}(1) \le a \text{ or } \bar{\phi}(1) \ge b} I(\phi)$, and we have used the upper bound implied by the aforementioned LDP in the final line. For this to be useful, we need to check that $J(a,b) > 0$. To this end, let $\mathcal{I} : H_0^1 \to C_{V_0}([0,1])$ be the Itô map which takes h to the solution to $\phi(t) = V_0 + \int_0^t c_\alpha \nu (t-s)^{H-\frac{1}{2}} \dot{h}_s \sigma(\phi(s)) ds$ (recall this solution is unique since σ is bounded and Lipschitz). Then

$$\begin{aligned}
|\mathcal{I}(h)(t) - V_0| \;=\; |\phi(t) - V_0| \;=\; \Big| \int_0^t c_\alpha \nu (t-s)^{H-\frac{1}{2}} \sigma(\phi(s)) \dot{h}_s ds \Big| \;&\le\; \|\sigma\|_\infty \int_0^t c_\alpha \nu (t-s)^{H-\frac{1}{2}} |\dot{h}_s| ds \\
&\le\; const. \times \frac{1}{2} \Big(\int_0^t \dot{h}_s^2 ds \Big)^{\frac{1}{2}} \\
&\le\; const. \times I(\phi)^{\frac{1}{2}}
\end{aligned}$$

where we have used Cauchy-Schwarz in the final line. If $J(a,b) = 0$ then for all $\varepsilon > 0$ there exists a $\phi_\varepsilon \in C_{V_0}([0,1])$ with $\underline{\phi}_\varepsilon(1) \le a$ or $\bar{\phi}_\varepsilon(1) \ge b$ and $I(\phi_\varepsilon) \le \varepsilon$, which (from the bound immediately above) implies that the sup norm of $\phi_\varepsilon - V_0$ is less than $(b - V_0) \wedge (V_0 - a)$ for ε sufficiently small, which is a contradiction. Thus $J(a,b) > 0$.

Similarly

$$\begin{aligned}
\mathbb{E}((e^{z\sqrt{t}} - S_t)^+) \;\ge\; \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+ 1_A) \;&=\; \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+) - \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+ 1_{A^c}) \\
&\ge\; \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+) \,-\, e^{z\sqrt{t}} e^{-\frac{1}{t^{2H}}(J(a,b) - \varepsilon)} .
\end{aligned}$$

Thus

$$\begin{aligned}
\frac{1}{\sqrt{t}} \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+) \,-\, \frac{e^{z\sqrt{t}}}{\sqrt{t}} e^{-\frac{1}{t^{2H}}(J(a,b) - \varepsilon)} \;&\le\; \frac{1}{\sqrt{t}} \mathbb{E}((e^{z\sqrt{t}} - S_t)^+) \\
&\le\; \frac{1}{\sqrt{t}} \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+) \,+\, \frac{e^{z\sqrt{t}}}{\sqrt{t}} e^{-\frac{1}{t^{2H}}(J(a,b) - \varepsilon)} .
\end{aligned}$$

$\hat{S}$ satisfies the conditions of the main Theorem 2.1 in [FSV21], so we have a small-t expansion for $\frac{1}{\sqrt{t}} \mathbb{E}((e^{z\sqrt{t}} - \hat{S}_t)^+)$, and for t small, $\frac{1}{\sqrt{t}} e^{z\sqrt{t}} e^{-\frac{1}{t^{2H}}(J(a,b) - \varepsilon)}$ is higher order than the error term in the main Theorem 2.1 in [FSV21], so the same expansion holds for $\frac{1}{\sqrt{t}} \mathbb{E}((e^{z\sqrt{t}} - S_t)^+)$.